FEDERATED LEARNING PHYSICAL AI ONCOLOGY TRIALS UNIFICATION

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ABSTRACT

The transition from conventional software-only artificial intelligence to physical AI systems incorporating robotic hardware in oncology clinical trials represents a paradigm shift requiring unified infrastructure for privacy, regulation, cross-framework interoperability, and multi-organization cooperation. This paper presents the PAI Oncology Trial FL platform (v1.1.0), a comprehensive federated learning framework comprising 235 Python modules (∼86,800 lines of code) that unifies five critical infrastructure pillars: (1) Privacy Infrastructure implementing all 18 HIPAA Safe Harbor identifiers with HMAC-SHA256 pseudonymization, (2) Regulatory Infrastructure spanning FDA, IRB, ICH-GCP, and multi-jurisdiction compliance across v0.6.0 and v0.9.1, (3) Cross-Framework Unification bridging NVIDIA Isaac Sim, MuJoCo, Gazebo, and PyBullet simulation environments, (4) Standards & Benchmarking for Q1 2026 objectives including model conversion and registry pipelines, and (5) Multi-Organization Cooperation enabling federated training across academic medical centers, community hospitals, and pharmaceutical companies. End-to-end workflow demonstrations are presented across 31 example scripts, 6 agentic AI production examples implementing Model Context Protocol (MCP), ReAct reasoning, real-time monitoring, autonomous orchestration, safety-constrained execution, and RAG-based compliance. A triple AI peer review process (v0.9.4–v0.9.9) using sequential Codex-to-Claude Code review-fix cycles resolved 31/31 code recommendations at 100% completion, establishing a dual-manufacturer trust benchmark for AI-generated clinical trial software. The platform demonstrates that unified federated learning infrastructure is a necessary precondition for transitioning the oncology industry to using robots in physical AI clinical trials.

Keywords Federated Learning · Physical AI · Oncology Clinical Trials · Digital Twins · Regulatory Compliance · HIPAA · Cross-Framework Unification · Agentic AI · Model Context Protocol · Differential Privacy

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1 Introduction

1.1 Current AI in Federated Learning Oncology

Federated learning (FL) has emerged as the dominant paradigm for privacy-preserving collaborative machine learning in oncology, enabling multiple institutions to jointly train AI models without exchanging raw patient data [1, 2]. Leading cancer centers including Dana-Farber have launched federated AI platforms where models travel to each center's secure data environment [3], and the European DigiONE consortium has demonstrated privacy-preserving learning health systems in precision oncology [4].

Current FL approaches in oncology typically address isolated aspects of the clinical trial workflow:

- **Model training:** FedAvg [1], FedProx [5], and SCAFFOLD [6] for heterogeneous hospital data
- **Privacy:** Differential privacy with Gaussian noise mechanisms [7] and secure aggregation protocols
- **Brain tumor segmentation:** Multi-institutional FL for glioblastoma without patient data sharing [8]

However, no existing open-source platform unifies federated learning, physical AI robotics, digital twins, regulatory compliance, privacy infrastructure, and clinical analytics into a single coherent framework for oncology clinical trials [9].

1.2 Benefits of Upgrading to AI Robots in Oncology Trials

The integration of physical AI—embodied robotic systems with onboard intelligence—into oncology clinical trials offers transformative benefits:

1. **Precision in tumor biopsies and drug delivery:** Surgical robots executing procedures with sub-millimeter accuracy guided by digital twin predictions, as modeled in the PAI platform's `physical_ai/` modules (v0.1.0)
2. **Consistent monitoring:** Autonomous robotic systems performing 24/7 patient monitoring with real-time anomaly detection via multi-modal sensor fusion (v0.3.0)
3. **Cross-site standardization:** Unified robot programming across trial sites eliminates procedural variability, enabled by the cross-framework unification layer (v0.3.0)
4. **Federated model improvement:** Robotic telemetry from multiple sites feeds back into federated model training without raw data exchange, as demonstrated in `01_mcp_oncology_server.py` (v0.5.0)

1.3 Projected Physical AI Benefits from Quantitative Codebase Data

The PAI Oncology Trial FL platform (v1.1.0) provides quantitative evidence for physical AI readiness:

Table 1: Platform quantitative metrics supporting physical AI oncology trial readiness

Metric	Value	Version
Total Python modules	235	v1.0.1
Lines of code	∼86,800	v1.0.1
Test files	82	v0.8.0+
Test directories	15	v0.8.0+
Example scripts	31	v0.5.0
Agentic AI examples	6	v0.5.0
CLI tools	5	v0.4.0
Security audit findings resolved	61	v0.7.0
Peer review recommendations resolved	31/31 (100%)	v0.9.4–v0.9.9
Regulatory frameworks supported	9	v0.9.1
HIPAA Safe Harbor identifiers	18/18	v0.6.0
Simulation frameworks bridged	4	v0.3.0
Tumor growth models	4	v0.4.0
Supported Python versions	3.10, 3.11, 3.12	v0.3.0+

The upstream Physical AI Oncology Trials repository [10] introduced the Unification Standard Level (USL) scoring framework evaluating 9 robotic systems across humanoid robots, surgical robots, and collaborative robots, with scores ranging from 3.4 (UFACTORY xArm7) to 7.4 (Franka Panda) [11]. The PAI Oncology Trial FL platform extends this foundation with federated learning capabilities necessary for multi-site robot deployment.

2 Methods

2.1 AI Development Tools and Models

The PAI Oncology Trial FL platform was developed using the following AI systems:

- **Primary development:** Anthropic Claude Code powered by Claude Opus 4.6 [12]—responsible for all code generation, documentation, and fix implementation across v0.1.0 through v1.1.0
- **Independent code review:** OpenAI ChatGPT-5.2-Codex [13]—performed three sequential senior engineering review cycles (v0.9.4, v0.9.6, v0.9.8) generating 31 total recommendations
- **Paper generation:** This paper was primarily generated by Claude Code (Opus 4.6) based on Mr. Kawchak's handwritten prompt, which took approximately 2 hours to compose

2.2 Development Timeline and Version History

The codebase was developed over the period February 17–27, 2026, with the repository converted from private to public on February 27, 2026. Table 2 summarizes the complete version history.

Table 2: Complete release timeline with AI contributor roles

Version	Date	AI Role	Key Deliverable
v0.1.0	2026-02-17	Claude Code	Initial FL platform
v0.2.0	2026-02-17	Claude Code	FedProx, SCAFFOLD, data harmonization
v0.3.0	2026-02-17	Claude Code	Unification framework, CI/CD
v0.4.0	2026-02-17	Claude Code	Digital twins, 5 CLI tools
v0.5.0	2026-02-17	Claude Code	31 example scripts
v0.6.0	2026-02-18	Claude Code	Privacy & regulatory (9 modules)
v0.7.0	2026-02-18	Claude Code	Security audit (61 findings)
v0.8.0	2026-02-18	Claude Code	Test suite (82 files, 15 dirs)
v0.9.0	2026-02-18	Claude Code	Clinical analytics (7 modules)
v0.9.1	2026-02-18	Claude Code	Regulatory submissions (6 modules)
v0.9.2	2026-02-18	Claude Code	Release documentation
v0.9.3	2026-02-19	Claude Code	Visualization (30 charts)
v0.9.4	2026-02-19	Codex	Peer review #1: 12 recommendations
v0.9.5	2026-02-19	Claude Code	Fix cycle #1: 12/12 resolved
v0.9.6	2026-02-19	Codex	Peer review #2: 9 recommendations
v0.9.7	2026-02-19	Claude Code	Fix cycle #2: 9/9 resolved
v0.9.8	2026-02-19	Codex	Peer review #3: 10 recommendations
v0.9.9	2026-02-19	Claude Code	Fix cycle #3: 10/10 resolved
v1.0.0	2026-02-19	Claude Code	Stable release
v1.0.1	2026-02-19	Claude Code	Documentation update
v1.1.0	2026-02-27	Claude Code	This paper and archival

Development statistics:

- **Total span:** 11 days (Feb 17–27, 2026)
- **Days with AI code updates:** 4 days (Feb 17, 18, 19, 27)
- **Days of AI code recommendations (Codex):** 1 day (Feb 19—v0.9.4, v0.9.6, v0.9.8)
- **Days of AI code fixes (Claude Code):** 4 days (Feb 17, 18, 19, 27—v0.1.0 through v1.1.0)
- **Human prompt authoring (Kevin Kawchak):** 18 prompts across development lifecycle, including a 2-hour handwritten prompt for this paper
- **Time from first prompt to PR submission:** Approximately 30 minutes (prompt submission: 2026-02-27 03:15 UTC, PR creation: 2026-02-27 03:45 UTC est.)

2.3 Prompt-Driven Development Methodology

All 18 development prompts are recorded in `prompts.md` [9]. The methodology follows a single-prompt delivery pattern where each prompt generates a complete, self-contained release. Key prompts include:

- **Prompt 1 (v0.1.0):** "Integrate physical AI with federated learning platform instructions into `pai-oncology-trial-fl`. Utilize `physical-ai-oncology-trials` for physical AI implementation."
- **Prompt 8 (v0.8.0):** "Build comprehensive pytest-based test suite achieving full coverage of all Python modules."
- **Prompt 13a (v0.9.4):** "Provide comprehensive code reviews to find errors across the entire codebase" (Codex)
- **Prompt 16 (v1.0.0):** "Prove and push a v1.0.0 stable release by analyzing the codebase end-to-end"

3 Results

3.1 Platform Architecture Overview

The PAI Oncology Trial FL platform follows a layered architecture where five infrastructure pillars support the end-to-end clinical trial workflow. Table 3 maps each pillar to its codebase location, version of introduction, and quantitative scope.

Table 3: Five-pillar architecture with codebase metrics

#	Pillar	Directory	LOC	Version
1	Privacy Infrastructure	`privacy/`	∼5,400	v0.6.0
2	Regulatory Infrastructure	`regulatory/`, `reg-sub/`	∼10,000	v0.6.0, v0.9.1
3	Cross-Framework Unification	`unification/`	∼8,200	v0.3.0
4	Standards & Benchmarking	`q1-2026-standards/`	∼3,800	v0.3.0
5	Multi-Org Cooperation	`federated/`	∼12,000	v0.1.0–v0.2.0
	Supporting: Clinical Analytics	`clinical-analytics/`	∼4,900	v0.9.0
	Supporting: Digital Twins	`digital-twins/`	∼4,500	v0.4.0
	Supporting: Physical AI	`physical_ai/`	∼6,200	v0.1.0
	Supporting: Agentic AI	`agentic-ai/`	∼6,700	v0.5.0
	Infrastructure: Tests	`tests/`	∼15,000	v0.8.0
	Infrastructure: CLI Tools	`tools/`	∼5,100	v0.4.0
	Infrastructure: Utilities	`utils/`, `scripts/`	∼500	v0.9.7–v0.9.9

```
PAI Oncology Trial FL - End-to-End Architecture (v1.1.0)
========================================================

+-- Multi-Organization Cooperation (Pillar 5) ----------+
|  FedAvg | FedProx | SCAFFOLD | Secure Aggregation     |
|  Site Enrollment | Data Harmonization | DP Budget      |
+-------+------+------+------+------+------+------------+
        |      |      |      |      |
+-------+--+ +-+------+-+ +--+------+--+ +-+----------+
| Privacy  | | Regulatory | | Unification | | Standards |
| (Pillar  | | (Pillar 2) | | (Pillar 3)  | | (Pillar  |
|  1)      | |            | |             | |  4)       |
| PHI Det. | | FDA Track  | | Isaac-MuJo  | | Model     |
| De-ID    | | IRB Mgmt   | | Bridge      | | Registry  |
| RBAC     | | GCP Check  | | Agent Unif. | | Benchmark |
| Breach   | | eCTD Comp. | | Format Conv | | Convert   |
+----+-----+ +-----+------+ +------+------+ +-----+----+
     |             |               |               |
+----+-------------+---------------+---------------+----+
|        Supporting Platform Modules                     |
| Digital Twins | Clinical Analytics | Agentic AI | CLI  |
| PK/PD | Survival | Risk | DSMB | MCP | ReAct | RAG   |
+-------------------------------------------------------+
        |
```

```
+-------+-----------------------------------------------+
|              Tests & Quality Infrastructure            |
| 82 test files | 15 directories | Security Audit       |
| Triple AI Peer Review | CI (3.10/3.11/3.12)           |
+-------------------------------------------------------+
```

3.2 Pillar 1: Privacy Infrastructure (v0.6.0)

The privacy framework implements comprehensive HIPAA Safe Harbor compliance across five modules totaling ∼5,400 LOC within the `privacy/` directory.

3.2.1 PHI/PII Detection and Management

The `phi_detector.py` module (v0.6.0) implements all 18 HIPAA Safe Harbor identifiers per 45 CFR 164.514(b)(2) [14]:

Listing 1: PHI detection categories from `privacy/phi-pii-management/phi_detector.py` (v0.6.0)

```
class PHICategory(Enum):
    NAME = "name"
    GEOGRAPHIC = "geographic_subdivision"
    DATE = "dates"
    PHONE = "phone_number"
    FAX = "fax_number"
    EMAIL = "email_address"
    SSN = "social_security_number"
    MEDICAL_RECORD = "medical_record_number"
    HEALTH_PLAN = "health_plan_beneficiary"
    ACCOUNT = "account_number"
    LICENSE = "certificate_license"
    VEHICLE = "vehicle_identifier"
    DEVICE = "device_identifier"
    URL = "web_url"
    IP_ADDRESS = "ip_address"
    BIOMETRIC = "biometric_identifier"
    PHOTO = "full_face_photo"
    OTHER = "other_unique_identifier"
```

3.2.2 De-identification Pipeline

The de-identification module provides six transformation methods with HMAC-SHA256 pseudonymization using cryptographically random salts (`os.urandom(32)`), as hardened during peer review cycle 3 (v0.9.8/v0.9.9):

- **REDACT:** Complete value removal
- **HASH:** HMAC-SHA256 pseudonymization (environment-driven keys post-v0.9.9)
- **GENERALIZE:** Geographic data to 3-digit ZIP codes
- **SUPPRESS:** Selective field suppression
- **PSEUDONYMIZE:** Consistent patient-level token generation
- **DATE_SHIFT:** Consistent patient-level date offsets

3.2.3 Access Control and Audit

Role-based access control (RBAC) implements 21 CFR Part 11 compliant audit trails [15] with five role tiers, time-limited access with fail-closed expiration (v0.6.0), and immutable audit log copies returned by `get_audit_log()` (v0.7.0 security fix).

3.2.4 Breach Response and Data Use Agreements

The breach response protocol implements four-factor risk assessment per 45 CFR 164.402 with 60-day notification tracking. The DUA generator supports five agreement types (research, clinical, vendor, multi-institutional, pediatric) with security tiers and configurable retention policies.

3.3 Pillar 2: Regulatory Infrastructure (v0.6.0, v0.9.1)

3.3.1 Core Regulatory Framework (v0.6.0)

The `regulatory/` directory implements five compliance domains:

Table 4: Regulatory infrastructure modules and compliance alignment

Module	Version	Standards
`fda_submission_tracker.py`	v0.6.0	FDA 510(k), De Novo, PMA
`irb_protocol_manager.py`	v0.6.0	ICH E6(R3), 21 CFR Part 11
`gcp_compliance_checker.py`	v0.6.0	ICH E6(R3) 13 principles
`regulatory_tracker.py`	v0.6.0	FDA, EMA, PMDA, TGA, HC
`HUMAN_OVERSIGHT_QMS.md`	v0.6.0	CRF risk tiers, AE boundaries

The FDA submission tracker supports five regulatory pathways—510(k), De Novo, PMA, Breakthrough, and Pre-Submission—referencing FDA AI/ML Device Guidance (January 2025) [16], PCCP Guidance (August 2025) [17], and QMSR (February 2026). The GCP compliance checker scores 13 ICH E6(R3) principles [18] excluding `NOT_ASSESSED` findings from the denominator (v0.7.0 logic fix).

3.3.2 Regulatory Submissions Platform (v0.9.1)

The `regulatory-submissions/` platform (∼4,600 LOC across 6 modules) complements clinical analytics with downstream regulatory packaging:

1. **Submission Orchestrator:** Lifecycle management for FDA submission workflows
2. **eCTD Compiler:** Module structure generation per FDA Technical Conformance Guide with SHA-256 checksums
3. **Compliance Validator:** Multi-regulation validation across 9 frameworks (HIPAA, GDPR, FDA 21 CFR 820, Part 11, ISO 14971, IEC 62304, ICH E6(R3)/E9(R1), MDR 2017/745)
4. **Document Generator:** Automated CSR, SAP, PCCP, ISO 14971, IEC 62304 documents
5. **Regulatory Intelligence:** 7-jurisdiction guidance tracking (FDA, EMA, PMDA, TGA, Health Canada, MHRA, NMPA)
6. **Submission Analytics:** KPIs, trend analysis, FDA MDUFA benchmarking

3.4 Pillar 3: Cross-Framework Unification (v0.3.0)

The `unification/` directory (v0.3.0) enables seamless interoperability across four simulation frameworks for surgical robotics:

```
Cross-Framework Unification Architecture (v0.3.0)
=================================================

+------------------+       +------------------+
| NVIDIA Isaac Sim |<----->|     MuJoCo       |
| (High-fidelity)  |       |  (Fast physics)  |
+--------+---------+       +--------+---------+
         |    Bidirectional Bridge   |
         +----------+  +------------+
                    |  |
             +------+--+------+
             | Unified State   |
             | Representation  |
             +------+--+------+
                    |  |
         +----------+  +------------+
         |                           |
+--------+---------+       +--------+---------+
|     Gazebo       |       |    PyBullet      |
|   (ROS 2)        |       | (Lightweight)    |
+------------------+       +------------------+
```

Key unification components include:

- **Isaac-MuJoCo Bridge** (`simulation_physics/`): Bidirectional state conversion with parameter mapping, enabling policy transfer between high-fidelity and fast physics engines
- **Unified Agent Interface** (`agentic_generative_ai/`): Tool format conversion for CrewAI, LangGraph, AutoGen, and custom backends
- **Cross-Platform Tools** (`cross_platform_tools/`): Framework detection (Isaac Lab, MuJoCo, Gazebo, PyBullet), model format conversion (URDF, MJCF, SDF, USD), and policy export with validation
- **Standards Protocols** (`standards_protocols/`): Data format standards, communication protocols, and safety compliance

3.5 Pillar 4: Standards & Benchmarking (v0.3.0)

The `q1-2026-standards/` directory implements three Q1 2026 objectives:

1. **Objective 1—Model Conversion Pipeline:** Cross-framework model format conversion with validation, supporting URDF↔MJCF↔SDF↔USD bidirectional transforms
2. **Objective 2—Model Registry:** Federated model registry with validation gates, version tracking, and compliance metadata
3. **Objective 3—Cross-Platform Benchmarking:** Standardized benchmark runner enabling reproducible comparison across simulation frameworks

The implementation guide includes a timeline (`timeline.md`) and compliance checklist (`compliance_checklist.md`) aligned with IEC 80601-2-77 [19] and ISO 14971 [20].

3.6 Pillar 5: Multi-Organization Cooperation (v0.1.0–v1.0.1)

The federated architecture enables multi-organization cooperation across the oncology trial ecosystem:

Table 5: Multi-organization cooperation model with platform integration points

Organization	Integration Points	Version
Academic Medical Centers	Federated training, IRB, digital twins	v0.1.0+
Community Hospitals	Site enrollment, data harmonization	v0.2.0+
Pharmaceutical Cos.	Treatment simulation, regulatory	v0.6.0+
Robotics Manufacturers	Simulation frameworks, surgical tasks	v0.3.0+
Regulatory Bodies	Submission tracking, compliance	v0.9.1+
Standards Organizations	IEC/ISO/ICH alignment	v0.3.0+

The federated learning core (`federated/`) implements three aggregation strategies with privacy guarantees:

- **FedAvg** (v0.1.0) [1]: Standard federated averaging for IID data distributions
- **FedProx** (v0.2.0) [5]: Proximal term regularization for non-IID heterogeneous hospital data
- **SCAFFOLD** (v0.2.0) [6]: Control variate correction for client drift across sites
- **Differential Privacy** (v0.1.0) [7]: Gaussian mechanism with gradient clipping and privacy budget (ε, δ) tracking
- **Secure Aggregation** (v0.1.0): Mask-based protocol ensuring no raw model weights are exchanged

3.7 Workflow Demonstrations: Domain Examples (v0.5.0)

Version 0.5.0 introduced 31 production-quality example scripts across five domains, demonstrating end-to-end workflow capabilities. These examples serve as reference implementations that clinicians, engineers, and regulators can use to evaluate the platform.

3.7.1 Core Federated Learning Examples

Five core examples (`examples/`, 4,924 LOC total) demonstrate progressive complexity:

1. `01_federated_training_workflow.py` (881 LOC): Multi-site FedAvg/FedProx pipeline with DP integration, secure aggregation, and convergence monitoring
2. `02_digital_twin_planning.py` (924 LOC): Patient digital twin construction with exponential, logistic, Gompertz, and von Bertalanffy tumor growth models
3. `03_cross_framework_validation.py` (865 LOC): Policy validation across Isaac Sim, MuJoCo, PyBullet, and Gazebo with kinematic drift analysis
4. `04_agentic_clinical_workflow.py` (917 LOC): Multi-agent clinical decision support with oncologist, radiologist, safety, and data agents
5. `05_outcome_prediction_pipeline.py` (982 LOC): Federated outcome prediction with survival analysis, biomarker integration, and fairness auditing

3.7.2 Agentic AI Examples—Production Architecture

Six agentic AI examples (`agentic-ai/examples-agentic-ai/`, ∼6,700 LOC total) provide production-grade implementations:

Table 6: Agentic AI example scripts with architecture patterns (v0.5.0)

#	Script	Architecture Pattern
01	`mcp_oncology_server.py`	Model Context Protocol server with tool registration, patient lookup, treatment simulation, protocol compliance, robotic telemetry
02	`react_treatment_planner.py`	ReAct reasoning loops with thought-action-observation cycles, 10+ step traces
03	`realtime_adaptive_monitoring.py`	Streaming multi-modal data, adaptive thresholds, CTCAE v5.0 grading
04	`autonomous_simulation_orchestrator.py`	Multi-framework job coordination with Pareto optimization
05	`safety_constrained_executor.py`	IEC 80601-2-77 safety gates, human-in-the-loop, emergency stop
06	`oncology_rag_compliance.py`	RAG with semantic chunking, cross-encoder reranking, citation tracking

MCP Oncology Server (`01_mcp_oncology_server.py`). This script implements a Model Context Protocol server [21] exposing five oncology domain tools for LLM agents:

Listing 2: MCP tool registration pattern from `01_mcp_oncology_server.py` (v0.5.0)

```
# Tool definitions with JSON Schema input validation
TOOLS = [
    {"name": "patient_lookup",
     "description": "Retrieve de-identified patient record",
     "inputSchema": {"type": "object",
       "properties": {"patient_id": {"type": "string"}}}},
    {"name": "treatment_simulation",
     "description": "Run digital twin treatment simulation",
     "inputSchema": {"type": "object",
       "properties": {"patient_id": {"type": "string"},
                      "modality": {"type": "string"}}}},
    {"name": "protocol_compliance",
     "description": "Verify ICH E6(R3) compliance",
     "inputSchema": {"type": "object",
       "properties": {"protocol_id": {"type": "string"}}}},
]
```

The MCP server integrates with the broader platform by:

- Querying the `privacy/` de-identification pipeline before returning patient records
- Invoking the `digital-twins/` patient modeling engine for treatment simulation
- Validating against the `regulatory/` GCP compliance checker for protocol verification
- Logging all tool invocations to 21 CFR Part 11 compliant audit trails with hash-chain integrity
- Ingesting robotic procedure telemetry from physical AI surgical systems

Synergistic inter-script operation. The six agentic AI scripts function both independently and synergistically:

- Scripts 01 (MCP) + 05 (Safety) combine to provide safety-constrained tool execution for LLM agents operating robotic systems
- Scripts 02 (ReAct) + 06 (RAG) combine reasoning with regulatory evidence grounding
- Scripts 03 (Monitoring) + 04 (Orchestration) enable real-time adaptive simulation campaigns where monitoring alerts trigger simulation parameter adjustments
- All six scripts share the platform's audit logging infrastructure, differential privacy protections, and HIPAA-compliant data handling

3.7.3 Physical AI Examples

Six physical AI examples (`examples-physical-ai/`, 800+ LOC each) demonstrate robotic oncology capabilities: real-time safety monitoring with ISO 13482 alignment, multi-sensor Kalman fusion, ROS 2 surgical deployment with lifecycle nodes, hand-eye calibration via Tsai-Lenz, shared autonomy teleoperation with variable blending, and robotic sample handling with chain-of-custody tracking.

3.8 Triple AI Peer Review Pipeline (v0.9.4–v0.9.9)

The triple AI peer review process constitutes a *sequential* pipeline of review-fix cycles alternating between two AI manufacturers, establishing a dual-manufacturer trust benchmark for AI-generated code [22]:

```
Sequential Triple AI Peer Review (v0.9.4 - v0.9.9)
==================================================

Claude Code --> Codex ------> Claude Code --> Codex
(codebase)     (review #1)    (fixes #1)     (review #2)
 v0.1-0.9.3     v0.9.4         v0.9.5         v0.9.6
                 12 recs        12/12          9 recs

--> Claude Code --> Codex ------> Claude Code
    (fixes #2)     (review #3)    (fixes #3)
     v0.9.7         v0.9.8         v0.9.9
     9/9            10 recs        10/10

Total: 31/31 recommendations resolved (100%)
```

This was **not** a parallel process. Each cycle was sequential: Claude Code provided the codebase, Codex reviewed and generated recommendations, Claude Code implemented all fixes, then Codex reviewed the updated codebase again. This sequential alternation—Claude Code → Codex → Claude Code → Codex → Claude Code → Codex—ensures each review cycle builds on the fixes from the previous cycle, enabling progressively deeper risk detection.

Table 7: Triple AI peer review cycle metrics (v0.9.4–v0.9.9)

Cycle	Review	Fixes	Recs	Resolved	Focus
1	v0.9.4 (Codex)	v0.9.5 (Claude)	12	12/12	CI & process
2	v0.9.6 (Codex)	v0.9.7 (Claude)	9	9/9	Compliance & security
3	v0.9.8 (Codex)	v0.9.9 (Claude)	10	10/10	Hardening & secrets
Total	**3 reviews**	**3 fix cycles**	**31**	**31/31**	**100%**

3.8.1 v1.0.1: Claude Code + Codex—31/31 Resolved as AI Bias Mitigation

The 31/31 resolution rate using two different AI manufacturers (Anthropic for code generation/fixes, OpenAI for independent review) serves as an example of AI bias mitigation in software development. By ensuring neither manufacturer's blind spots persist in the final codebase, the dual-manufacturer approach establishes an emerging software quality methodology:

- **Manufacturer A (Anthropic Claude Code):** May have systematic blind spots in security patterns, exception handling, or compliance areas

- **Manufacturer B (OpenAI Codex):** Independent review catches issues that Manufacturer A's training data may not flag
- **Result:** The alternating sequential process eliminates single-manufacturer bias, producing code that has been independently validated by two separate AI architectures

3.9 Code Trust and Safety: Tests, Checks, and Migrations

3.9.1 Core Test Suite (v0.8.0)

The test suite at `tests/` (v0.8.0) contains 82 test files across 15 directories, constituting a core infrastructure for code trust:

Table 8: Test suite directory structure (v0.8.0, 15 directories)

Directory	Files	Coverage Scope
`test_federated/`	6	Coordinator, client, aggregation, DP
`test_privacy/`	6	PHI, de-identification, RBAC, breach
`test_regulatory/`	5	FDA, IRB, GCP, intelligence, oversight
`test_clinical_analytics/`	7	Orchestration, PK/PD, survival, DSMB
`test_regulatory_submissions/`	4	eCTD, compliance, documents, analytics
`test_agentic_ai/`	5	MCP, ReAct, monitoring, orchestrator
`test_digital_twins/`	3	Patient modeling, treatment, integration
`test_physical_ai/`	4	Digital twin, robotics, sensor fusion
`test_unification/`	3	Bridge, agent interface, converter
`test_tools/`	2	CLI tool validation
`test_standards/`	3	Q1 2026 conversion, registry, benchmark
`test_examples/`	3	Example script execution
`test_regression/`	2	v0.7.0 audit finding regression guards
`test_integration/`	2	End-to-end cross-module pipelines
`conftest.py`	1	Shared fixtures, `np.random.seed(42)`
Total	**82** (incl. 16 `__init__.py`)	

3.9.2 Security Audit (v0.7.0)

The comprehensive security audit (v0.7.0) identified and resolved 61 findings across three categories:

- **12 Security vulnerabilities:** `torch.load` → `weights_only=True`, `np.load` → `allow_pickle=False`, plaintext key removal, mutable audit log exposure
- **14 Logic bugs:** Division-by-zero guards, bounded `while` loops with `MAX_ITERATIONS`, floating-point tolerance (`np.isclose`), hazard ratio sign correction
- **35 Compliance findings:** `RESEARCH USE ONLY` disclaimers in all modules, audit trail logging, date shift handling, `model_type="unspecified"` defaults

3.9.3 Peer Review Fixes and Migrations

Cycle 1 (v0.9.4→v0.9.5)—CI Determinism:

- Created `constraints/lint.txt` with pinned tool versions for CI reproducibility
- Added `.pre-commit-config.yaml` for local lint consistency
- 5 files modified, 4 new files, +67 net LOC

Cycle 2 (v0.9.6→v0.9.7)—Compliance & Security:

- Fixed 24 UTC timestamp instances across 23 files (migrated to `datetime.timezone.utc`)
- Narrowed 17 broad exception handlers to specific exception types
- Replaced MD5 hashing with SHA-256
- Created `scripts/check_version_alignment.py` (71 LOC), `scripts/security_scan.py` (96 LOC), `scripts/release_metrics.py` (89 LOC)
- 23 files modified, 3 new files, +25 net LOC

Cycle 3 (v0.9.8→v0.9.9)—Hardening & Secrets:

- Removed hardcoded cryptographic keys from 8 modules (migrated to `os.environ` via `utils/crypto.py`)
- Implemented audit integrity chain verification
- Created PHI-safe logging sanitizer (`utils/log_sanitizer.py`, 36 LOC)
- Narrowed 11 additional exception handlers
- Created centralized configuration (`utils/config.py`, 66 LOC) and structured error codes (`utils/error_codes.py`, 74 LOC)
- Added PHI fuzz tests (`tests/test_privacy/test_phi_fuzz.py`, 127 LOC)
- 22 files modified, 7 new files, +386 net LOC

Cumulative impact across all trust & safety activities:

- 92 total issues resolved (61 audit + 31 peer review)
- 28 exception handlers narrowed across 19 files
- 25 UTC timestamp fixes
- 13 new utility scripts/modules
- +478 net lines of trust infrastructure code

3.10 Clinical Analytics Platform (v0.9.0)

The `clinical-analytics/` platform (∼4,900 LOC across 7 modules) provides privacy-preserving analytics for multi-site oncology trials:

- **PK/PD Engine:** One- and two-compartment pharmacokinetic models with analytical and ODE solutions (`scipy.integrate.solve_ivp`), NCA metrics (AUC, C_{max}, T_{max}, $t_{1/2}$), E_{max} dose-response
- **Survival Analysis:** Kaplan-Meier with Greenwood SE, log-rank test, Cox proportional hazards via BFGS with Breslow ties, Harrell's C-index, restricted mean survival time (RMST)
- **Risk Stratification:** Composite weighted scores, adaptive enrichment with per-stratum decision support, Hosmer-Lemeshow calibration
- **Clinical Interoperability:** ICD-10/SNOMED CT crosswalk, LOINC, RxNorm, MedDRA mappings, CDISC SDTM export [23]
- **Consortium Reporting:** DSMB reports with SHA-256 integrity hashing for tamper detection

3.11 Digital Twins for Oncology (v0.4.0)

The `digital-twins/` platform implements patient-specific modeling across three subdirectories:

- **Patient Modeling** (`patient_model.py`): Six tumor types with four growth kinetics models—exponential ($V(t) = V_0 e^{kt}$), logistic ($V(t) = \frac{K}{1+(\frac{K}{V_0}-1)e^{-rt}}$), Gompertz ($V(t) = K \cdot (\frac{V_0}{K})^{e^{-\alpha t}}$), and von Bertalanffy
- **Treatment Simulation** (`treatment_simulator.py`): Multi-modality treatment including chemotherapy, radiation, immunotherapy, targeted therapy, and combination protocols with RECIST 1.1 response assessment [24]
- **Clinical Integration** (`clinical_integrator.py`): EHR, PACS, LIS, RIS, and TPS system integration with a federated clinical bridge

The visualization infrastructure (v0.9.3) provides 30 Plotly-based interactive charts across three thematic sets for publication-quality rendering of digital twin data.

3.12 Technology Stack and Dependencies

The platform is designed for minimal dependency footprint while supporting optional advanced capabilities. Table 9 summarizes the technology stack across required and optional tiers.

Table 9: Technology stack with version requirements and integration status

Package	Version	Purpose	Status
Required (Python 3.10+, no GPU)			
NumPy	≥1.24.0	Array computation, federated MLP	Integrated
SciPy	≥1.11.0	ODE solvers, optimization	Integrated
scikit-learn	≥1.3.0	ML algorithms	Integrated
pandas	≥2.0.0	Data manipulation	Integrated
cryptography	≥41.0.0	HMAC-SHA256, encryption	Integrated
PyYAML	≥6.0	Configuration parsing	Integrated
pytest	≥7.4.0	Testing framework	Integrated
ruff	==0.15.1	Linting & formatting (pinned)	CI-enforced
Optional: Deep Learning & Medical Imaging			
PyTorch	≥2.5.0	Deep learning models	Bridge ready
MONAI	≥1.3.0	Medical image analysis	Optional
pydicom	≥2.4.0	DICOM parsing	Optional
Optional: Agentic AI Frameworks			
LangChain	≥0.3.0	LLM chains	Interface
CrewAI	≥0.80.0	Multi-agent orchestration	Interface
MCP	≥1.1.0	Model Context Protocol	Interface
Optional: Simulation Frameworks			
NVIDIA Isaac Sim	4.2.0	High-fidelity surgical sim	Bridge
MuJoCo	≥3.1.0	Physics simulation	Bridge
Gazebo	Harmonic	ROS 2 integration	Guide
PyBullet	≥3.2.6	Lightweight simulation	Guide

3.13 Federated Coordinator Implementation Details

The core federated learning coordinator (`federated/coordinator.py`, v0.2.0) implements three aggregation strategies with convergence detection. The following code illustrates the FedProx proximal term, which penalizes client models that deviate too far from the global model—critical for non-IID hospital data distributions:

Listing 3: FedProx aggregation with proximal regularization from `federated/coordinator.py` (v0.2.0)

```
class FederatedCoordinator:
    """Orchestrates federated training rounds with
    FedAvg, FedProx, or SCAFFOLD aggregation."""

    def aggregate_fedprox(self, client_updates, global_weights):
        """FedProx: weighted average with proximal term mu."""
        total_samples = sum(u["n_samples"] for u in client_updates)
        aggregated = {}
        for key in global_weights:
            weighted_sum = sum(
                u["weights"][key] * (u["n_samples"] / total_samples)
                for u in client_updates
            )
            # Proximal term: mu * (w - w_global)
            proximal = self.mu * (weighted_sum - global_weights[key])
            aggregated[key] = weighted_sum - proximal
        return aggregated
```

The data harmonization module (`federated/data_harmonization.py`, v0.2.0, ∼10,161 bytes) provides DICOM/FHIR vocabulary mapping and unit conversion across hospital sites, ensuring that heterogeneous data formats are standardized before federated aggregation. The site enrollment manager (`federated/site_enrollment.py`, v0.2.0, ∼11,523 bytes) implements quality-weighted site selection for multi-site trial coordination.

3.14 CLI Tools for Trial Operations (v0.4.0)

Five command-line tools provide operational support for trial engineers:

1. **DICOM Inspector** (`tools/dicom-inspector/`): Inspection and validation of DICOM medical images with tag verification and JSON output
2. **Dose Calculator** (`tools/dose-calculator/`): Radiation and chemotherapy dose calculation with NCCN/RTOG reference constants and BSA-based dosing

3. **Trial Site Monitor** (`tools/trial-site-monitor/`): Multi-site monitoring with GREEN/YELLOW/RED status classification and enrollment tracking
4. **Sim Job Runner** (`tools/sim-job-runner/`): Cross-framework simulation job launcher with automatic framework detection
5. **Deployment Readiness** (`tools/deployment-readiness/`): Checklist-based validation scoring with PASS/FAIL/WARNING/REQUIRES_VERIFICATION status levels

4 Discussion

4.1 Unified Platforms as a Precondition for Physical AI Adoption

The PAI Oncology Trial FL platform demonstrates that unified infrastructure—combining federated learning, privacy, regulation, cross-framework interoperability, and multi-organization cooperation—is a *necessary precondition* for transitioning the oncology industry to using robots in physical AI clinical trials. Without such unification:

- **Privacy fragmentation** would require each trial site to independently implement HIPAA Safe Harbor de-identification, increasing the risk of compliance gaps
- **Regulatory inconsistency** would delay FDA submissions as each site uses different tracking systems with incompatible audit trails
- **Simulation incompatibility** would prevent cross-site validation of robotic procedures, as policies trained in Isaac Sim cannot directly execute in MuJoCo or Gazebo without the unification bridge
- **Aggregation vulnerability** would expose federated models to data poisoning without standardized secure aggregation and differential privacy across all participating sites

The platform's 235 Python modules provide a single coherent answer to these fragmentation risks, enabling the industry transition from software-only AI to physical AI in oncology trials.

4.2 Triple AI Peer Review as Quality Assurance Methodology

The sequential Codex-to-Claude Code peer review pipeline (v0.9.4–v0.9.9) establishes a replicable methodology for AI code quality assurance. The correction trajectory—Cycle 1 (CI/process) $\rightarrow$ Cycle 2 (compliance/security) $\rightarrow$ Cycle 3 (hardening/secrets)—demonstrates systematic risk elimination where each cycle addresses progressively deeper issues. All six review/fix releases were completed on 2026-02-19, a pace impractical for human reviewers processing 86,800 LOC [22].

The dual-manufacturer approach mitigates single-AI bias: Codex (OpenAI) reviewing Claude Code (Anthropic) ensures neither manufacturer's training data blind spots persist in the final codebase. This is analogous to dual-auditor requirements in financial compliance, where independence between auditors strengthens assurance.

4.3 Agentic AI for Clinical Trial Operations

The six agentic AI examples (v0.5.0) demonstrate that LLM agents can be safely integrated into clinical trial workflows when constrained by:

1. **Safety gates** (Script 05): IEC 80601-2-77 aligned pre/post-condition checks with human-in-the-loop approval for high-risk actions [19]
2. **Protocol grounding** (Script 06): RAG-based retrieval from regulatory documents with faithfulness verification and citation tracking
3. **Audit trails** (Script 01): 21 CFR Part 11 compliant logging of all LLM tool invocations with hash-chain integrity [15]
4. **Privacy enforcement** (All scripts): PHI detection and de-identification applied before any data leaves the local site

4.4 Comparison with Existing Approaches

Compared to existing federated learning platforms for oncology (Dana-Farber multi-center platform [3], DigiONE [4]), PAI Oncology Trial FL is unique in integrating:

- Physical AI robotics (surgical robot interfaces, sensor fusion, cross-framework simulation)
- Complete regulatory submission pipeline (eCTD compilation, multi-jurisdiction intelligence)
- Agentic AI capabilities (MCP, ReAct, RAG for clinical trial operations)
- Digital twins with four tumor growth kinetics models

- Triple AI peer review for code quality assurance

4.5 Quantitative Development Velocity

The development velocity achieved by AI-driven development is quantifiable from the release history:

Table 10: AI development velocity metrics by date

Date	Releases	Est. LOC Added	Key Achievements
2026-02-17	v0.1.0–v0.5.0	∼55,000	Core platform + 31 examples
2026-02-18	v0.6.0–v0.9.1	∼25,000	Privacy, security, analytics
2026-02-19	v0.9.2–v1.0.1	∼6,800	Peer review, visualization, docs
2026-02-27	v1.1.0	∼500	Paper, documentation updates
Total	**21 releases**	**∼86,800**	**4 calendar days of AI work**

This represents approximately 21,700 LOC per active development day—a rate enabled by Claude Code Opus 4.6's ability to generate production-quality Python with comprehensive docstrings, type annotations, test coverage, and regulatory compliance disclaimers in each prompt cycle. The PAI Oncology Trial FL platform builds upon the original Physical AI Oncology Trials repository [10] (51 Python modules, ∼40,526 LOC), extending it with a 4.3× increase in codebase size and the addition of federated learning, clinical analytics, regulatory submissions, and the triple AI peer review quality process.

5 Limitations and Future Work

5.1 Current Limitations

5.1.1 Claude Code Limitations

- **Context window constraints:** Large codebases require careful prompt decomposition; the 18-prompt development strategy was necessary because no single prompt could address the full 86,800 LOC platform
- **Deterministic output variability:** Despite `np.random.seed(42)` seeding, Claude Code's stochastic generation may produce slightly different implementations across identical prompts
- **Domain knowledge boundaries:** Clinical oncology details (e.g., specific drug dosing protocols, exact FDA submission timelines) required external verification by Mr. Kawchak
- **Testing limitations:** While Claude Code generated 82 test files, property-based testing coverage (via Hypothesis) and integration test depth could be expanded

5.1.2 Codex Limitations

- **Review-only role:** Codex was restricted to generating recommendations without directly modifying code, limiting its ability to verify fix correctness
- **Recommendation granularity:** Some recommendations (e.g., "improve exception handling") required Claude Code to interpret scope and priority
- **Contextual blind spots:** Codex reviews focused on individual files rather than cross-module architectural patterns, potentially missing systemic issues

5.1.3 Human Oversight Limitations

- **Prompt engineering effort:** Mr. Kawchak authored 18 prompts over the development lifecycle; scaling this approach requires prompt engineering expertise
- **Review bottleneck:** All AI-generated PRs required human review and merge approval, creating a serial bottleneck in the development pipeline
- **Clinical validation gap:** The platform is research-grade only; clinical deployment requires independent IRB approval, FDA clearance, and prospective clinical validation

5.2 Future Work

Three development proposals are documented in `DEVELOPMENT_PROPOSALS.md` [9]:

1. **Real-Time Federated Safety Signal Detection:** Pharmacovigilance algorithms (PRR, ROR, BCPNN, MGPS, SPRT), MedDRA classification, ICH E2B reporting
2. **Genomic Biomarker Integration Pipeline:** VCF/MAF variant processing, ACMG/AMP classification, TMB/MSI/HRD scoring, federated GWAS with privacy preservation
3. **Adaptive Trial Design Engine:** Bayesian inference, response-adaptive randomization, Thompson sampling, platform/master protocol designs

6 Conclusion

This paper presents the PAI Oncology Trial FL platform (v1.1.0)—a comprehensive federated learning framework for physical AI oncology trials comprising 235 Python modules (∼86,800 LOC) developed across 21 releases over 11 days. The platform unifies five critical infrastructure pillars: privacy (18 HIPAA Safe Harbor identifiers), regulation (FDA/IRB/GCP/multi-jurisdiction), cross-framework interoperability (Isaac Sim/MuJoCo/Gazebo/PyBullet), standards and benchmarking, and multi-organization cooperation.

End-to-end workflow demonstrations across 31 example scripts—including six production-grade agentic AI implementations using MCP, ReAct, real-time monitoring, autonomous orchestration, safety-constrained execution, and RAG-based compliance—establish that the platform supports the full clinical trial lifecycle from federated model training through regulatory submission.

The triple AI peer review process (v0.9.4–v0.9.9) resolved 31/31 code recommendations at 100% completion through sequential Codex-to-Claude Code review-fix cycles, demonstrating that dual-manufacturer AI code review can operate at speeds and depths impractical for traditional human-only review of codebases exceeding 86,000 LOC. Combined with the 61 security audit findings resolved in v0.7.0, the platform has undergone 92 total quality corrections establishing a verifiable trust chain.

Unified platforms such as PAI Oncology Trial FL are a necessary precondition for transitioning the oncology industry to using robots in physical AI clinical trials. The open-source availability of this platform (MIT license) enables the research community to build upon, validate, and extend these capabilities toward clinical deployment.

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The author of the article declares no competing interests.

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